

Unit 11 (Part 2)

The Eigenvalue Decomposition

Unit Goals

- Be able to define what is meant by an eigenvalue of a matrix.
- Be able to solve for the eigenvalues of 2-by-2 and 3-by-3 matrices.
- Be able to solve for the eigenvalues of certain n -by- n matrices with characteristic forms.
- Be able define what is meant by an eigenvector and an eigenspace.
- Be able to find a linearly independent set of eigenvectors associated with a given eigenvalue of a matrix.
- Be able to diagonalize a matrix, or determine that it can not be diagonalized, given the eigenvalues.
- Be able to use the eigenvalue decomposition to solve real-world problems.

Unit 11.2

The Eigenvalue Decomposition

Eigenvalues and eigenvectors

$$A\vec{x} = \lambda\vec{x}$$

An $n \times n$ matrix has at least 1 and at most n eigenvalues:

- Solutions to $\det(\lambda I - A) = 0$ where λ is a variable.
- For each eigenvalue λ_i , the null space of $(\lambda_i I - A)$ is the eigenspace associated with λ_i ; λ_i is a constant.
- The eigenspace is characterized by a basis; at least one vector for each distinct eigenvalue.

The eigenvectors of different eigenvalues are linearly independent

- Let $\vec{x}_1$, $\vec{x}_2$ and $\vec{x}_3$ be any three eigenvectors of A , and
- Let λ_1 , λ_2 and λ_3 be their corresponding eigenvalues:
$$A\vec{x}_1 = \lambda_1\vec{x}_1, \quad A\vec{x}_2 = \lambda_2\vec{x}_2, \quad A\vec{x}_3 = \lambda_3\vec{x}_3$$
- If $\vec{x}_2$ is a scalar multiple of $\vec{x}_1$ they must have the same eigenvalue:

$$\begin{aligned} \vec{x}_2 = k\vec{x}_1 &\rightarrow \\ A\vec{x}_2 = A(k\vec{x}_1) &= k(A\vec{x}_1) = k(\lambda_1\vec{x}_1) = \lambda_1(k\vec{x}_1) = \lambda_1\vec{x}_2 \end{aligned}$$

Compare to $A\vec{x}_2 = \lambda_2\vec{x}_2$, we obtain:

$$\lambda_1 = \lambda_2$$

The eigenvectors of different eigenvalues are linearly independent

$$A\vec{x}_1 = \lambda_1\vec{x}_1, \quad A\vec{x}_2 = \lambda_2\vec{x}_2, \quad A\vec{x}_3 = \lambda_3\vec{x}_3$$

- If $\vec{x}_1$, $\vec{x}_2$ and $\vec{x}_3$ are linearly dependent but not scalar multiples of each other, they must all have the same eigenvalue.

$$\vec{x}_3 = k\vec{x}_1 + l\vec{x}_2 \rightarrow$$

$$A\vec{x}_3 = kA\vec{x}_1 + lA\vec{x}_2 \rightarrow$$

$$k\lambda_1\vec{x}_1 + l\lambda_2\vec{x}_2 = \lambda_3\vec{x}_3$$

$$k\lambda_1\vec{x}_1 + l\lambda_2\vec{x}_2 = k\lambda_3\vec{x}_1 + l\lambda_3\vec{x}_2$$

$$k(\lambda_1 - \lambda_3)\vec{x}_1 + l(\lambda_2 - \lambda_3)\vec{x}_2 = \vec{0}$$

$$\rightarrow \lambda_1 = \lambda_2 = \lambda_3$$

The case of n distinct eigenvalues.

If an $n \times n$ matrix has n distinct eigenvalues:

- Each eigenvalue must generate at least one linearly independent eigenvector.
- We must have at least n linearly independent eigenvectors in total.
- We can not have more than n linearly independent eigenvectors.
- Each eigenvalue must generate exactly one linearly independent eigenvector.
- The set generated by taking one eigenvector for each eigenvalue is a basis for $\mathbb{R}^n$.

The case of n distinct eigenvalues.

- Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be distinct eigenvalues for an $n \times n$ matrix A , and let $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ be any corresponding eigenvectors.
- We can express the eigenvalue equation in matrix form:
$$A\vec{x}_i = \lambda_i\vec{x}_i, \forall i \in [1, n] \rightarrow AX = X\Lambda$$
- X is a matrix whose columns are the eigenvectors.
- Λ is a diagonal matrix whose entries are the corresponding eigenvalues.

$$X = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n], \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix}$$

The case of n distinct eigenvalues.

$$\begin{aligned} AX &= A[\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n] \\ &= [A\vec{x}_1 \quad A\vec{x}_2 \quad \cdots \quad A\vec{x}_n] \\ &= [\lambda_1\vec{x}_1 \quad \lambda_2\vec{x}_2 \quad \cdots \quad \lambda_n\vec{x}_n] \end{aligned}$$

$$X\Lambda = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix} = [\lambda_1\vec{x}_1 \quad \lambda_2\vec{x}_2 \quad \cdots \quad \lambda_n\vec{x}_n]$$

$$A\vec{x}_i = \lambda_i\vec{x}_i, \forall i \in [1, n] \rightarrow AX = X\Lambda$$

The case of n distinct eigenvalues.

$$A\vec{x}_i = \lambda_1 \vec{x}_i, \quad \forall i \in [1, n] \rightarrow AX = X\Lambda$$

- X is a matrix whose columns are the eigenvectors; it will always be nonsingular as the columns must be linearly independent.
- Λ is a diagonal matrix whose entries are the corresponding eigenvalues; it will be nonsingular if and only if all the eigenvalues are nonzero.

$$AX = X\Lambda \rightarrow A = X\Lambda X^{-1}, \quad \Lambda = X^{-1}AX$$

The Eigenvalue decomposition:

$$A = X\Lambda X^{-1}$$

The case of repeated eigenvalues.

- In some cases an $n \times n$ matrix A , does not have n distinct eigenvalues; some of the eigenvalues are “repeated” when we factor $\det(\lambda I - A)$.
- The repeated eigenvalues may generate more than one linearly independent eigenvector or may not!

The case of repeated eigenvalues.

Definitions:

- The **geometric multiplicity** of an eigenvalue is the dimension of the associated eigenspace
 - The number of linearly independent eigenvectors we can find
- The **algebraic multiplicity** of an eigenvalue is number of times it is repeated when we factor the characteristic polynomial.
 - For e.g. $(\lambda_1 - 2)^2 (\lambda_2 - 1)^3$, λ_1 repeats two times and λ_2 repeats three times.

The case of repeated eigenvalues.

- **Example:** $A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 \\ 1 & \lambda - 2 \end{vmatrix} = \lambda(\lambda - 2) - (-1) = (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$$

$$\lambda = 1 \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = R \rightarrow \vec{x}_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace associated with $\lambda = 1$.
- **There is a single eigenvalue with an algebraic multiplicity of 2 but a geometric multiplicity of 1.**

The case of repeated eigenvalues.

- **Example:** $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

$$|\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 \\ 0 & \lambda - 3 \end{vmatrix} = (\lambda - 3)^2 = 0 \rightarrow \lambda = 3$$

$$\lambda = 3 \rightarrow (\lambda I - A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = R \rightarrow \vec{x}_{n1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace associated with $\lambda = 3$.
- **There is one eigenvalue with an algebraic and geometric multiplicity of 2.**
- **The eigenvectors span the entire $\mathbb{R}^2$ space.**

The case of repeated eigenvalues.

- If an $n \times n$ matrix A , has n linearly independent eigenvectors...
- Let $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ be such a set, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be corresponding eigenvalues.
- We can express the eigenvalue equation in matrix form:
$$A\vec{x}_i = \lambda_i\vec{x}_i, \forall i \in [1, n] \rightarrow AX = X\Lambda$$

The case of repeated eigenvalues.

- X is a matrix whose columns are the eigenvectors.
- Λ is a diagonal matrix whose entries are the corresponding eigenvalues.

$$X = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n], \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix}$$

- **Note: If any eigenvalue has a geometric multiplicity of more than 1, it will be repeated in the matrix Λ**

The Eigenvalue Decomposition

- Any $n \times n$ matrix A with n linearly independent eigenvectors can be expressed as:

$$A = X\Lambda X^{-1}$$

- where X is a nonsingular matrix whose columns are the eigenvectors;
- and Λ is a diagonal matrix whose entries are the corresponding eigenvalues.
- A matrix that has a well-defined eigenvalue decomposition is said to be **diagonalizable**.
- Performing an eigenvalue decomposition is sometimes called **diagonalizing** the matrix.

The utility of the EVD: A^k

$$A = X\Lambda X^{-1} \rightarrow A^k = \prod_{i=1}^k A = AA \cdots A = (X\Lambda X^{-1})(X\Lambda X^{-1}) \cdots (X\Lambda X^{-1})$$

$$A^k = X \left(\prod_{i=1}^k \Lambda \right) X^{-1} = X\Lambda^k X^{-1}$$

$$\Lambda^k = \left(\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix} \right)^k = \begin{bmatrix} (\lambda_1)^k & 0 & \cdots & 0 \\ 0 & (\lambda_2)^k & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & (\lambda_n)^k \end{bmatrix}$$

The utility of the EVD: $A\vec{x}_i = \vec{x}_{i+1}$

$$A = X\Lambda X^{-1}, \quad A\vec{x}_i = \vec{x}_{i+1}$$

$$\rightarrow XA\vec{x}_i = X\vec{x}_{i+1}$$

$$\rightarrow XAX^{-1}X\vec{x}_i = X\vec{x}_{i+1}$$

$$\rightarrow \Lambda X\vec{x}_i = X\vec{x}_{i+1}$$

$$A\vec{x}_i = \vec{x}_{i+1} \rightarrow \Lambda\hat{x}_i = \hat{x}_{i+1}, \quad \hat{x}_i = X\vec{x}_i$$

$$\Lambda\vec{u} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} \lambda_1 u_1 \\ \lambda_2 u_2 \\ \vdots \\ \lambda_n u_n \end{bmatrix}$$

Unit 11.2.1

**Worked examples: Solving
problems with the Eigenvalue
Decomposition**

Nondiagonalizable matrices.

- Demonstrate why it is impossible to find a nonsingular X and diagonal Λ such that $A = X\Lambda X^{-1}$ where:

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

Nondiagonalizable matrices.

- Demonstrate why it is impossible to find a nonsingular X and diagonal Λ such that $A = X\Lambda X^{-1}$ where:

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 2 & -3 & -4 \\ 0 & \lambda - 2 & -3 \\ 0 & 0 & \lambda - 2 \end{vmatrix}$$

$$= (\lambda - 2)(\lambda - 2)(\lambda - 2)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 2)^3 = 0 \rightarrow \lambda = 2$$

There is a single eigenvalue with an algebraic multiplicity of 3.

Nondiagonalizable matrices.

- Demonstrate why it is impossible to find a nonsingular X and diagonal Λ such that $A = X\Lambda X^{-1}$ where:

$$\lambda = 2, A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 0 & -3 & -4 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -3 & -4 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_n = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

- The eigenspace of the only eigenvalue has dimension of 1 (which is less than 3); we can not find three linearly independent **eigenvectors!**

Unit 11.3

The Spectral Decomposition

The spectral theorem

Real symmetric matrices, $A^T = A$:

- Are always diagonalizable: all eigenvalues have a geometric multiplicity equal to the algebraic multiplicity.
- Always have eigenvalues that are exclusively real numbers.
- Always have eigenvectors corresponding to different eigenvalues that are orthogonal!
 - We can always construct an orthonormal set of eigenvectors!

$$A = X\Lambda X^{-1} \rightarrow A = Q\Lambda Q^T, Q^T = Q^{-1}$$

Important: We can always write this form, but we need to construct orthonormal bases for each eigenspace: normalization only if the dimension is 1, Gram-Schmidt otherwise!

Positive Definite Matrices

A symmetric matrix with **ONLY POSITIVE** eigenvalues is termed **positive definite**.

- For a **positive definite** matrix:

$$\vec{x}^T S \vec{x} > 0, \forall \vec{x} \neq \vec{0}$$

- Note, if all eigenvalues are simply **NON-NEGATIVE**, the matrix is **positive semi-definite**, and we have

$$\vec{x}^T S \vec{x} \geq 0, \forall \vec{x} \neq \vec{0}$$

- **Important application:** the extremum test in calculus.

The extremum test in Calculus.

- Local extrema (maxima or minima) of a (differentiable) function of multiple variable are located where all partial derivatives are equal to zero:

$$\nabla F(\vec{x}^*) = \left[\frac{\partial F(\vec{x}^*)}{\partial x_i} \right] = \vec{0}$$

- The Hessian matrix is a matrix of all the second partial derivatives:

$$H(\vec{x}^*) = \left[\frac{\partial^2 F(\vec{x}^*)}{\partial x_i \partial x_j} \right]$$

The extremum test in Calculus.

The Hessian is symmetric everywhere the second derivatives are continuous.

- The Hessian has a well-defined spectral decomposition: $H(\vec{x}^*) = Q\Lambda Q^T$
- If all eigenvalues are positive, the Hessian is **positive definite**:
 - The extremum is a local minimum.
- If all eigenvalues are negative, the Hessian is **negative definite**:
 - The extremum is a local maximum.

Find the spectral decomposition.

- If possible, find **orthogonal** Q and diagonal Λ such that where $A = Q\Lambda Q^T$.

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Find the spectral decomposition.

- If possible, find **orthogonal** Q and diagonal Λ such that where $A = Q\Lambda Q^T$.

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 & -1 \\ 0 & \lambda - 4 & 0 \\ -1 & 0 & \lambda - 3 \end{vmatrix}$$

$$= -0 + (\lambda - 4) \begin{vmatrix} \lambda - 3 & -1 \\ -1 & \lambda - 3 \end{vmatrix} - 0 = (\lambda - 4)((\lambda - 3)^2 - 1)$$

$$= (\lambda - 4)(\lambda^2 - 6\lambda + 9 - 1) = (\lambda - 4)(\lambda^2 - 6\lambda + 8) = (\lambda - 4)(\lambda - 4)(\lambda - 2)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 4)^2(\lambda - 2) = 0 \rightarrow \lambda = 2, 4$$

- There one eigenvalue with an algebraic multiplicity of 2 and one with a multiplicity of 1.

Find the spectral decomposition.

- If possible, find **orthogonal** Q and diagonal Λ such that where $A = Q\Lambda Q^T$.

$$\lambda = 2, A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 0 \\ -1 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_n = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

- We want an **orthonormal** basis:

$$\vec{q}_{\lambda=2} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Find the spectral decomposition.

- If possible, find **orthogonal** Q and diagonal Λ such that where $A = Q\Lambda Q^T$.

$$\lambda = 4, A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

- We want an **orthonormal** basis.

$$(\vec{x}_{n2})^T \vec{x}_{n1} = 0 \rightarrow \vec{q}_{\lambda=4,1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{q}_{\lambda=4,2} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Find the spectral decomposition.

- If possible, find **orthogonal** Q and diagonal Λ such that where $A = Q\Lambda Q^T$.

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow \lambda = 2, 4, \vec{q}_{\lambda=4,1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{q}_{\lambda=4,2} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \vec{q}_{\lambda=2} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Using the eigenvalue decomposition.

- Given the spectral decomposition:

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

- Find a general expression for $C = A^k$.
- Find a matrix such that $B^2 = A$.

Using the eigenvalue decomposition.

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

- Find a general expression for $C = A^k$.

$$\begin{aligned} C = A^k &= Q\Lambda^k Q^T = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 4^k & 0 & 0 \\ 0 & 4^k & 0 \\ 0 & 0 & 2^k \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 4^k & 0 \\ 4^k/\sqrt{2} & 0 & 4^k/\sqrt{2} \\ -2^k/\sqrt{2} & 0 & 2^k/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 4^k/2 + 2^k/2 & 0 & 4^k/2 - 2^k/2 \\ 0 & 4^k & 0 \\ 4^k/2 - 2^k/2 & 0 & 4^k/2 + 2^k/2 \end{bmatrix} \end{aligned}$$

Using the eigenvalue decomposition.

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

- Find a matrix such that $B^2 = A$. **Let $B = Q\Lambda^{1/2}Q^T$ then $B^2 = Q\Lambda Q^T = A$**

$$\begin{aligned} B = Q\Lambda^{1/2}Q^T &= \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 \\ \sqrt{2} & 0 & \sqrt{2} \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 + 1/\sqrt{2} & 0 & 1 - 1/\sqrt{2} \\ 0 & 2 & 0 \\ 1 - 1/\sqrt{2} & 0 & 1 + 1/\sqrt{2} \end{bmatrix} \end{aligned}$$